

part 2: SMC for High-Order integrators

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1 Homogeneity

Homogeneous control laws appear as solutions to many control problems such as a minimum time feedback control for the chain of integrators or the high-order sliding mode design. The homogeneity allows some time constraints in control systems to be fulfilled by means of a proper selection of the so-called homogeneity degree. Similar to the linear case, an asymptotic stability of a homogeneous system implies its robustness (input-to-state stability) with respect to a certain class of parametric uncertainties and exogenous perturbations.

Many literatures like (POLYAKOV, 2020; ~~XXX~~, 2010; SHITESSEL et al., 2014) talks about homogeneity.

Corollary 6.1 The global uniform finite-time stability of homogeneous differential equations (Filippov inclusions) with negative homogeneity degree is **robust with respect to locally small homogeneous perturbations**.

1.1 Homogeneity of coordinate, function and vector field

Assign a weight(the **homogeneity degree**) of each coordinate $x_i \in \mathbb{R}, i = 1, \dots, n$, where $m_i > 0$. We will write

$$\deg(x_i) = m_i. \quad (1)$$

The corresponding simple linear transformation

$$d_k : (x_1, x_2, \dots, x_n) \mapsto (\kappa^{m_1} x_1, \kappa^{m_2} x_2, \dots, \kappa^{m_n} x_n) \quad (2)$$

is called **homogeneity dilation**, and $\kappa > 0$ is called its parameter.

A **function** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called homogeneous of the degree (weight) $q \in \mathbb{R}$ with the above homogeneity dilation and written as $\deg(f) = q$, if for any $\kappa > 0$, the identity $f(d_k x) = \kappa^q f(x)$ holds.

A **vector field** $f : \mathbb{R}^n \rightarrow \mathbb{R}^n, f = (f_1, \dots, f_n)^T$, is called homogeneous of degree $q \in \mathbb{R}$ with the above dilation and written as $\deg(f) = q$, if all its componenets f_i are homogeneous and the identities

$$\deg f_i = \deg x_i + \deg f = \deg x_i + q, i = 1, 2, \dots, n \quad (3)$$

Let A and B be two homogeneous functions of $x \in \mathbb{R}^n$ different from identical zero, and let λ be a real number; then

1. The sum of A and B is a homogeneous function only if $\deg A = \deg B$
2. $\forall \lambda \neq 0$, we have $\deg \lambda = 0$
3. $\deg AB = \deg A + \deg B$
4. $\deg \frac{A}{B} = \deg A - \deg B$
5. $\deg \lambda A = \deg A$
6. $\deg \frac{\partial}{\partial x_i} A = \deg A - \deg x_i$ if $\frac{\partial}{\partial x_i} A$ is not identical zero

To verify the last equality it can be seen that

$$\begin{aligned} \frac{\partial}{\partial \kappa_i^m x_i} A(d_k x) &= \kappa^{-m_i} \frac{\partial}{\partial x_i} \kappa^{\deg A} A(x) \\ &= \kappa^{\deg A - m_i} \frac{\partial}{\partial x_i} A(x) \end{aligned} \quad (4)$$

The last equality tells that for a systme $\dot{x} = f(x)$, we have $\dot{x}_i = f_{i(x)}$. Then,

$$\deg \dot{x}_i = \deg f_i = \deg x_i - \deg t \quad (5)$$

1.2 Homogeneity of differential equations and inclusions

Take the one dimension system $\dot{x} = f(x) = x^2$ into consideration, let $\deg x = 1$, then the homogeneity degree of **function** f is $\deg f = 2$ and the homogeneity degree of **vector field** f is $\deg f = \deg f_i - \deg x_i = 1$. *The ambiguity disappears if we speak about the homogeneity of the differential equation $\dot{x} = f(x)$.*

We call differential equation $\dot{x} = f(x)$ homogeneous with degree q , if the system is invariant with respect to the linear time-coordinate transformation

$$G_k : (t, x) \mapsto (\kappa^{-1}t, d_k x), \kappa > 0. \quad (6)$$

Then we have

$$f(x) = \kappa^{-1} d_k^{-1} f(d_k x) \quad (7)$$

. The homogeneity degree of the system is $\deg t = \deg f_i - \deg x_i = \deg f$.

$$\begin{aligned} f_{i(d_k x)} &= \kappa^{\deg f_i} f_{i(x)} = \kappa^{\deg x_i + \deg f} f_{i(x)} \\ f(d_k x) &= d_k \kappa^{\deg f} f(x) \end{aligned} \quad (8)$$

Note: The nonzero homogeneity degree q of a vector field can always be scaled to ± 1 by an appropriate proportional change to the weights of the coordinates and time.

Definition 6.6 A vector-set field $F(x) \subset \mathbb{R}^n, x \in \mathbb{R}^n$, and the differential inclusion

$$\dot{x} \in F(x) \quad (9)$$

are called homogeneous of the degree $q \in \mathbb{R}$ with the dialtion, which is written as $\deg F = q$, if the DI is invariant with respect to the time-coordinate transformation (6)

1.3 Convergence Rates of Homogeneous Algorithms

1.3.1 Finite-Time and Fixed-Time Stabilization

Consider simplest scalar first order system

$$\dot{x} = u \quad (10)$$

- The classical approach gives the standard *linear* proportional feedback

$$u_{\text{linear}}(x) = -x \quad (11)$$

which guarantees an asymptotic(exponential) convergence to the origin of any trajectory of the closed-loop system: $|x(t)| = e^{-t}|x_0|$.

- The **globally homogeneous** feedback is

$$u_{\text{FT}}(x) = -\sqrt{|x|}\text{sign}(x). \quad (12)$$

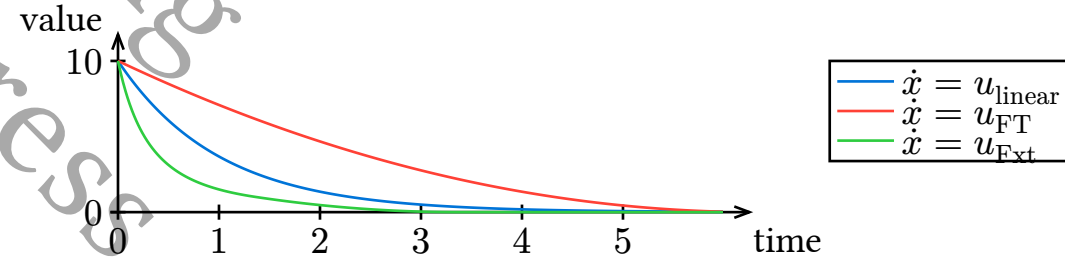
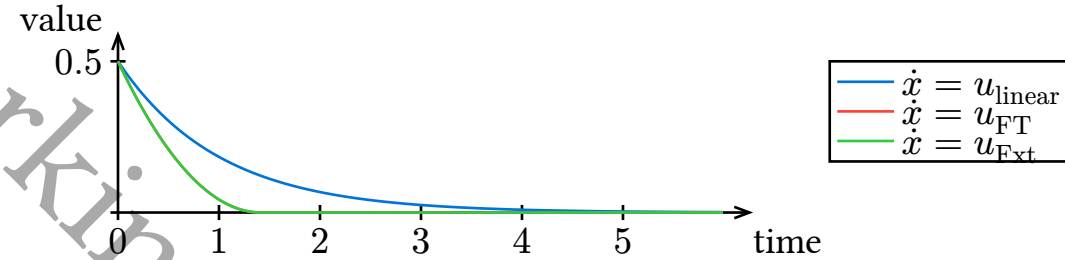
This algorithm stabilizes the system at the origin in a *finite time*:

$$x(t) = 0, \text{ for } t \geq T(x_0) = 2\sqrt{|x_0|} \quad (13)$$

- The *fixed-time* stabilizing controller can be selected **locally homogeneous** in the form:

$$u_{\text{FT}}(x) = \begin{cases} -[x]^{\frac{1}{2}} & |x| \leq 1 \\ -[x]^{\frac{3}{2}} & |x| > 1 \end{cases} \quad (14)$$

The system will be stabilized within 4 second, that is $x(t) = 0, \forall t \geq 4 \forall x_0 \in \mathbb{R}$



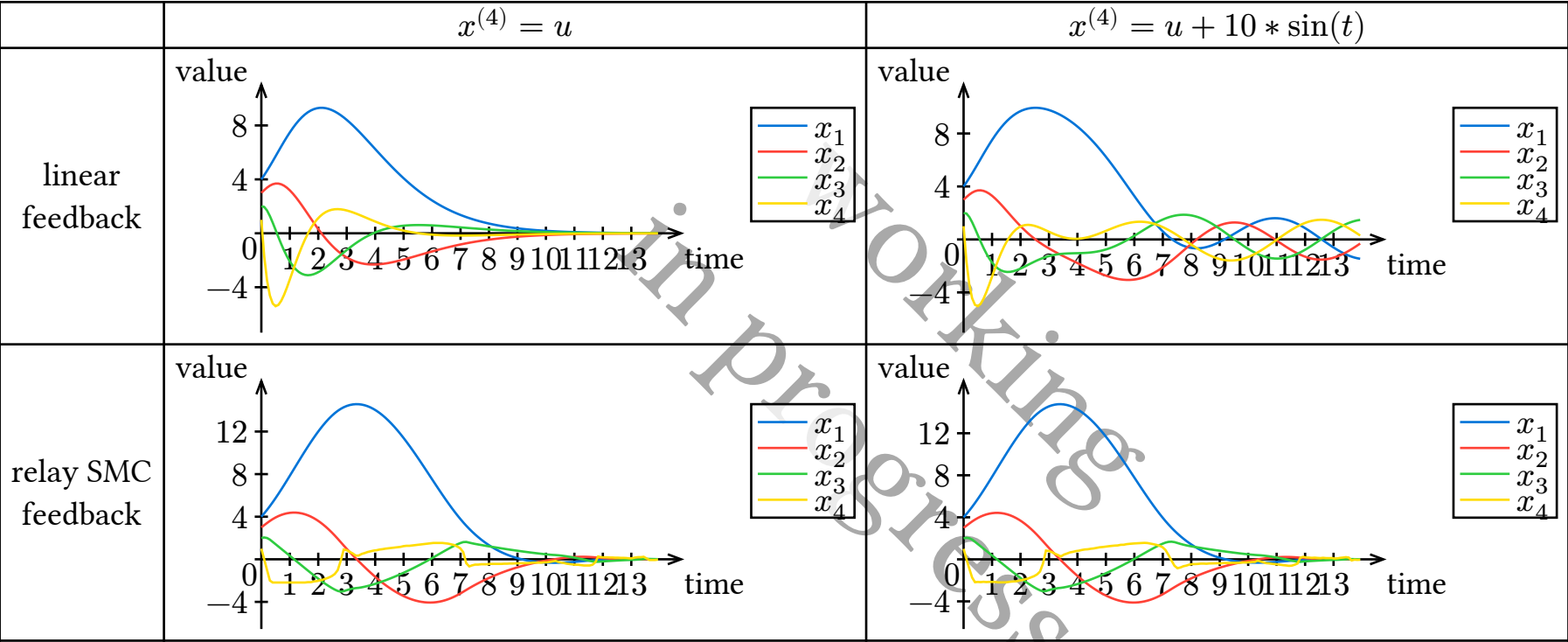
1.3.2 Robustness

1.3.3 Elimination of an Unbounded “Peaking” Effect

working
in progress

2 High-Order Sliding Mode Control for Integrator Systems

Traditionally, a linear feedback can stabilize high-order system without robustness



Consider the system

$$\sigma^{(r)} = u + \delta. \quad (15)$$

Nested Sliding Controllers are given by

$$\begin{aligned} u &= -\alpha \Psi_{r-1,r}(\sigma, \dot{\sigma}, \dots, \sigma^{(r-1)}) \\ \Psi_{0,r} &= \text{sign}(\sigma) \\ \Psi_{i,r} &= \text{sign}(\sigma^{(i)} + \beta_i N_{i,r} \Psi_{i-1,r}) \\ N_{i,r} &= \left(|\sigma|^{\frac{1}{r}} + |\dot{\sigma}|^{\frac{q}{r-1}} + \dots + |\sigma^{\frac{q}{r-i+1}}| \right)^{\frac{1}{q}} \end{aligned} \quad (16)$$

THANKS